HDOC- 8 – 1
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Program to Find the Largest Among Three Numbers Using if–else

This document provides a clean, well-commented implementation for finding the largest of three numbers in C. The program takes three integers as input and uses an if–else if–else ladder with relational and logical operators to decide and display the largest number.

1. C Implementation

Step 1: Understand the Logic

- A number is the largest if it is greater than or equal to both of the other two numbers.
- Check the first number against the other two; if it fails, check the second number.
- If neither the first nor the second is the largest, the third number must be the largest.
- The logical AND operator (&&) combines the two comparisons, and >= handles equal values.

Step 2: Define Variables and Data Types

- Use three integer variables (int a, b, c) to store the input numbers.

Step 3: Take User Input

- Display a prompt using printf("Enter three numbers: "), then read the values using scanf("%d %d %d", &a, &b, &c).

Step 4: Apply the if–else if–else Ladder

1. If a >= b and a >= c, then a is the largest.
2. Else if b >= a and b >= c, then b is the largest.
3. Otherwise (else), c is the largest.

Step 5: Format the Output

- Print the largest number along with a message using printf().

Complete C Code

```c
/*
 * Q11 (User Inputs, Operations & Output)
 * Write a program to input three numbers and find the
 * largest among them using if-else.
 */
#include <stdio.h>

int main() {
    int a, b, c;

    // Step 1: Read the three numbers
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    // Step 2: Compare the numbers using if-else
    if (a >= b && a >= c) {
        printf("%d is the largest number.\n", a);
    } else if (b >= a && b >= c) {
        printf("%d is the largest number.\n", b);
    } else {
        printf("%d is the largest number.\n", c);
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc8_1.c -o hdoc8_1
$ ./hdoc8_1
Enter three numbers: 10 25 15
25 is the largest number.
```

Run 2:

```
$ ./hdoc8_1
Enter three numbers: 42 7 19
42 is the largest number.
```

Run 3:

```
$ ./hdoc8_1
Enter three numbers: 3 8 90
90 is the largest number.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 10 25 15 (a = 10, b = 25, c = 15)
- Calculation:
  - a >= b and a >= c: 10 >= 25 is FALSE, so the condition is FALSE
  - b >= a and b >= c: 25 >= 10 and 25 >= 15 is TRUE, so the else if block runs
- Output: 25 is the largest number.

Sample Test Case 2:

- Input: 42 7 19 (a = 42, b = 7, c = 19)
- Calculation:
  - a >= b and a >= c: 42 >= 7 and 42 >= 19 is TRUE, so the first if block runs
- Output: 42 is the largest number.

Sample Test Case 3:

- Input: 3 8 90 (a = 3, b = 8, c = 90)
- Calculation:
  - a >= b and a >= c: 3 >= 8 is FALSE, so the condition is FALSE
  - b >= a and b >= c: 8 >= 3 is TRUE but 8 >= 90 is FALSE, so the condition is FALSE
  - The final else block runs, so c is the largest
- Output: 90 is the largest number.